

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/232510918>

Profiling the Differences in Students' Knowledge, Interest, and Strategic Processing

Article in *Journal of Educational Psychology* · September 1998

DOI: 10.1037/0022-0663.90.3.435

CITATIONS

217

READS

2,179

2 authors, including:



[Patricia Alexander](#)

University of Maryland, College Park

315 PUBLICATIONS 18,697 CITATIONS

SEE PROFILE

Profiling the Differences in Students' Knowledge, Interest, and Strategic Processing

Patricia A. Alexander and P. Karen Murphy
University of Maryland

The purpose for this research was threefold: to determine whether distinct and informative student profiles would emerge from knowledge, interest, and strategy measures specific to educational psychology; to compare these profiles with prior studies; and to explore changes in student profiles across an academic semester. As a result of cluster-analytic procedures, 3 distinct groups of participants emerged at pretest, and 4 emerged at posttest. One of the profiles that remained fairly consistent from pretest to posttest was the Learning-Oriented cluster. Students fitting this profile began the semester with the highest means in interest and strategic processing and with a moderate level of domain knowledge. By the end of the semester, this cluster obtained the highest means also on the domain knowledge test. Yet, the largest cluster at posttest seemed unable or unwilling to learn from demanding exposition.

In the last decade, the research on individual differences has moved toward the exploration of intraindividual variabilities (Ackerman, Kyllonen, & Roberts, in press). That is to say, researchers have become increasingly concerned with how cognitive and noncognitive factors, such as knowledge and self-regulation, configure differently *within* individuals (Ackerman, Kanfer, & Goff, 1995). Our purpose in this study was to explore such intraindividual differences by examining the various learner profiles that emerged within one classroom. Specifically, we wanted to ascertain whether undergraduates enrolled in an educational psychology course would naturally form into distinctive clusters based on their academic characteristics. We formed these clusters from information on students' subject matter knowledge, interests, and strategic processing. Because they were constructed by means of cluster analysis, the resulting profiles encompassed not only multiple indicators for each student but also data from every participating student.

In addition, we set out to compare the academic profiles in this study with other characterizations in the literature, to examine their consistency (e.g., de Jong & Ferguson-Hessler, 1986; Renkl, 1997). Surely, cognitive and noncognitive diversity exists in any classroom. If the clusterings we detected manifested similarities to profiles reported in related research, then their diagnostic and instructional relevance would seemingly be augmented. Thus, it was important to us not only whether such profiles could be statistically verified but also whether they would be instructionally valuable. Further, we wanted the opportunity to

inspect the changes that occurred in students over the semester as a way of probing the malleability of these intraindividual profiles. As educators, our assumption is that formal learning positively affects participating students. Tracking any shifts in student profiles from the beginning to the end of the semester would offer clues to the potential impact school-based experiences have on mature learners' knowledge, interests, and strategic processing.

Our expectation was that instruction over time should create greater diversity among our students in terms of their knowledge, interest, and strategic behavior, particularly because most students enter this learning environment as relative novices. For instance, one might predict that a clustering of students who begin instruction with some knowledge and interest in the content, and with the ability to perform strategically, would have an advantage over those who initially have little knowledge or interest in the subject matter, or who must struggle to learn new or demanding information (Alexander, 1997). Should such profiles exist, then the differences among them might well be magnified by a semester of study (Stanovich, 1986). A rise in the mean differences for knowledge, interest, and strategic processing among clusters from the outset to the conclusion of a course would be indicative of this type of fan-spread effect (Bryk & Weisberg, 1977). Equally interesting, the interaction between the cognitive and motivational measures might well vary among the different clusters from pretest to posttest. In essence, not only would the mean scores differ among clusters but the correlational patterns would vary as well.

We chose knowledge, interest, and strategic processing as our clustering variables in this study for several reasons. First, it can be argued that one of the central aims of schooling is the construction of principled knowledge about fundamental academic domains or disciplines (Alexander, Murphy, & Woods, 1996; Gelman & Greeno, 1989). We expect that students who participate in formal instruction for 12 or more years will garner a body of relevant and usable knowledge in such fields as mathematics, science, history, and reading. We use the term *domain knowledge* to refer to

Patricia A. Alexander and P. Karen Murphy, Department of Human Development, University of Maryland.

A version of this article was presented at the annual meeting of the American Educational Research Association, Chicago, April, 1997.

Correspondence concerning this article should be addressed to Patricia A. Alexander, Department of Human Development, University of Maryland, College Park, Maryland 20742-1131. Electronic mail may be sent to pa34@umail.umd.edu.

this corpus of knowledge specific to a particular field of study (Alexander, Schallert, & Hare, 1991).

Yet researchers have found that those who attain competence or reach expertise in any given field not only are knowledgeable but also manifest a strong personal interest or investment in that domain (e.g., Guthrie & Wigfield, 1997; Mitchell, 1993; Renninger, 1992). In addition, they are skilled in reasoning and problem solving (e.g., Alexander, Graham, & Harris, in press; Holyoak, 1991; Pirolli & Recker, 1994). Through our research on academic development, the perception that competence and expertise are better understood as a multidimensional interplay of knowledge, interest, and strategic components repeatedly has been reinforced (e.g., Alexander, 1997; Alexander, Kulikowich, & Schulze, 1994; Alexander, Murphy, Woods, Duhon, & Parker, 1997). Because of these presumed linkages, we used measures of personal or individual interest (Hidi, 1990; Schiefele, 1991) and strategic processing (Pressley, Borkowski, & Schneider, 1989; Pressley et al., 1992; Weinstein & Mayer, 1986), along with a domain knowledge task, to build our learner profiles in this study.

With this choice of clustering variables, what kind of profiles might we expect in a classroom of undergraduates studying educational psychology? Research on problem solving (e.g., Recker & Pirolli, 1995; Renkl, 1997) and expert-novice distinctions (e.g., Chi, Bassok, Lewis, Reimann, & Glaser, 1989; de Jong & Ferguson-Hessler, 1986) provided us clues as to the potential intraindividual differences we might anticipate. These cluster-analytic studies were conducted across various domains (e.g., physics, programming, and mathematics) and involved learners of differing ages and varied abilities. Thus, even though studies in these areas did not incorporate motivational measures and rarely included dedicated measures of strategic processing, they still produced some important and consistent groupings that informed the present investigation.

In particular, several studies identified important characteristics of more successful versus less successful performers. For instance, individuals who were generally successful at task performance either (a) had the benefit of a substantial body of relevant domain knowledge on which they relied or (b) appeared to overcome lower levels of domain knowledge through marked effort and effective strategic behavior (Recker & Pirolli, 1995). Renkl (1997) referred to the former group of successful performers as *anticipative reasoners* and the latter clustering as *principle-based explainers*.

In contrast, students who were novice or poor in their problem solving seemed to treat the given task or problem in a cursory manner. As a result, they failed to deal with its inherent structure or complexity (e.g., Chi et al., 1989; de Jong & Ferguson-Hessler, 1986, 1991). Renkl (1997) used the label *superficial* to depict this profile. Other less successful students appeared to recognize the limitations of their performance but apparently did not take advantage of available resources or did not use compensatory strategies. Students matching this particular profile were described as *passive learners* (Recker & Pirolli, 1995; Renkl, 1997). Thus, the research on problem solving and expert-novice differences suggests that students within classrooms can be

distinguished on the basis of the relevant knowledge they may or may not possess and of their ability and willingness to engage in the effortful and strategic processing of a challenging task.

In a recent study, Alexander, Jetton, and Kulilowich (1995) profiled students' academic learning in the field of human biology-immunology also by means of cluster analysis. Alexander et al. (1995) included a measure of student interest, along with gauges of domain knowledge and strategic processing. Participants included premedical students and students in an educational psychology graduate program (Experiment 1), as well as undergraduates enrolled in an introductory educational psychology class (Experiment 2). Several informative clusters, with similarities to those described, emerged across the two experiments. In general, we would characterize these as low, high, and mixed profiles. For instance, the low-profile group, who scored the lowest performance on the criterion measure (i.e., text comprehension task), knew little and reportedly cared little about the domain and had difficulties recalling what they had read. This low-profile group seems similar to the passive and superficial groups in the Recker and Pirolli (1995) and Renkl (1997) studies.

The high-profile cluster, by comparison, were (a) most knowledgeable in human biology-immunology, (b) most interested in that field, and (c) skilled at learning from demanding texts. This cluster has the most in common with the experts or good problem solvers in the Chi et al. (1989) and de Jong and Ferguson-Hessler (1986, 1991) research. Between these two clusters was a mixed profile, consisting of skillful novices who seemed rather capable of learning from demanding exposition but still knew little about human biology-immunology. These students were only moderately interested in the passage content. This group parallels the principle-based explainers in Renkl's (1997) study, who also overcame their initial deficits in knowledge to perform well on the task. In Experiment 2, Alexander et al. (1995) found another mixed cluster of undergraduates who had an adequate base of subject matter knowledge but did not perform particularly well when reading or recalling challenging subject matter text. According to the researchers, the poor performance of this particular cluster could be the result of either an inability or an unwillingness to apply appropriate text-processing strategies.

Thus, with these prior studies as a backdrop, we set out to extend the research in several ways. First, we chose a varied domain. Certainly, cluster analysis has shown promise in separating more successful from less successful learners in scientific and mathematical domains. However, this approach has not been as frequently applied in domains, like educational psychology, that are more ill structured (i.e., less procedurally rich or problem based) in nature (Alexander, 1992; Spiro, Coulson, Feltovich, & Anderson, 1994; Spiro & Jehng, 1990). Second, our intention was to allow domain knowledge to vary freely—neither to exacerbate its effects nor diminish the role it might play for these students. In prior studies, knowledge was often manipulated. For instance, certain researchers selected participants thought to be quite distinct in their knowledge base, as in the case of experts

versus novices (Chi et al., 1989), good versus poor problem solvers (de Jong & Ferguson-Hessler, 1986), or premedical versus non-premedical students (Alexander et al., 1995, Experiment 1). Conversely, other researchers used novel and challenging tasks they believed would lessen the impact of prior knowledge (Renkl, 1997). Finally, we set out to explore changes that might result in student profiles after formal instruction in the field of educational psychology. The prior studies that guided our thinking had not considered the possibility of shifts in learner profiles over time and, thus, left this important question unanswered.

Method

Participants

The participants for this study were 329 students enrolled in an introductory course in educational psychology taught by Alexander. The course was a requirement for their certification in elementary and secondary teacher education. The majority of these students (approximately 80%) were juniors and seniors at a large university in the Southwest. The proportion of women and men in the sample was typical for teacher education programs (i.e., 280 women and 49 men). Most participants (92%) were Caucasian, with a small percentage of Hispanic (6%) and African American (2%) students.

Clustering Measures

Because we were concerned with the ecological validity of the procedures that would result in the learner profiles, we administered the battery of measures described herein at the outset and the conclusion of the course under the following conditions. At pretest, students were informed that their performance on these measures would not be reflected in their course grade but would be used to modify the existing curriculum in accordance with their demonstrated strengths and weaknesses. At posttest, students' performance on the various knowledge, interest, and strategy tasks was part of their final examination. In this manner, we believed that the students would approach the tasks as relevant components of the course and not as experimental measures that were tangential to their learning and performance.

Domain knowledge measure. In constructing a measure of domain knowledge, we wanted an instrument that would afford us a picture of students' breadth of knowledge in the field of educational psychology versus their depth of understanding of selected concepts, procedures, or topics (cf. Alexander et al., 1994). We developed items from the instructional materials for the course and the required text (Goetz, Alexander, & Ash, 1992). The resulting measure of educational psychology consisted of 24 items. Sixteen items were multiple-choice questions, and 8 were commonality items (i.e., items for which students had to identify a superordinate term for four given terms). Examples are provided in prior research (e.g., Alexander, Murphy, Woods, et al., 1997). These items were dichotomously scored as correct or incorrect (highest possible score = 24). The Kuder-Richardson 20 for the measure at pretest was .62. We considered this to be an acceptable level of reliability, given the relatively low variability of scores ($M = 12.66$, $SD = 2.26$).

Interest scale. Our measure of student interest was a 26-item scale that tapped students' interest in concepts associated with the domain of educational psychology. Students indicated their interest in these concepts on a 7-point rating scale, ranging from 1 (*not very*

interested) to 7 (*very interested*). They were also given the option of indicating that they were "uncertain of the meaning" of a particular concept or topic. We considered the terms on the scale to be representative of central concepts within the domain (e.g., memory, problem solving, teacher-made tests, and learned helplessness). Moreover, these concepts were addressed within the course text and class discussions. The Cronbach alpha for the scale (highest possible score = 182) was .82.

Strategy measures. Our strategic processing dimension in the profiles was represented by two measures that the students completed in conjunction, a strategy-use inventory and a recall task. Specifically, students studied a given text and maintained a record of the strategies they used during the process (strategy-use inventory). Then, after studying, the students answered two questions about the passage content (recall task).

For both of these measures, we considered the choice of texts to be a significant factor. Not only did we want the content to be relevant to the course but we also wanted the texts to be sufficiently lengthy and challenging to warrant strategic effort. Using these criteria, we chose two passages from a review of research dealing with the Learner-Centered Psychological Principles (American Psychological Association, Presidential Task Force on Psychology in Education, 1993), which we had authored (Alexander & Murphy, 1998). One passage surveyed research on knowledge and its role in learning. The other was concerned with motivation and affect. The knowledge passage consisted of 1,025 words, 85 sentences, and 10 paragraphs. The motivation passage was 1,126 words, 93 sentences, and 9 paragraphs in length. Each student read one of the passages, presented in counterbalanced order, at pretest and the other at posttest. Before they read, we informed the students that they would answer questions about the passages when they completed their studying. They were then allowed to read and study the passages for as long as they wanted. All of the students finished this task within 30 min. After studying the passage and before the recall task, students were presented with an intervening task (i.e., long-division math problem).

The recall task consisted of two questions. The first question asked students to write down any important information they could remember from the passage. The second requested any additional information that they could recall. For the sake of this analysis, we combined students' answers to these two questions into one recall score. To aid us in scoring, we constructed a response protocol for each text. These protocols were essentially checklists containing gist statements for each of the sentences in the passages (85 for the knowledge passage and 93 for the motivation passage). For example, one sentence in the knowledge text read, "In essence, the knowledge base is the total of all an individual knows or believes" (Alexander et al., 1991). We indicated this sentence on the protocol with the phrase "knowledge base = all one knows/believes." When it came to scoring students' recalls, we checked their written responses against the protocol to gauge whether the information listed was (a) in the text and (b) an accurate rendering of the content. Even though some students included the same idea in their answers to both questions, we counted an entry only once. To determine the recall score, we simply tallied the number of marks received for the two questions.

In the past, we have developed response protocols in which each entry signified an idea unit roughly equivalent to a subordinate clause or phrase conveying a complete thought (Pichert & Anderson, 1977). However, given the length of the chosen texts and the general nature of the questions that we asked, we did not consider this level of specificity necessary for scoring students' recalls in this study. This perception was confirmed during scoring. Interrater reliability for scoring the recall task was .92, with any differences resolved in discussion. Also, because no students approached

ceiling for either passage, we did not consider the discrepancy in number of sentences per passage (85 vs. 93) to be a problem.

As they studied the texts, students were instructed to monitor their strategy use. We furnished a strategic-use inventory for this purpose. The inventory, which consisted of 15 general text-processing strategies, had been used in prior research ($\alpha = .80$; Knight & Alexander, 1994). We submitted the pretest data to a common factor analysis to extract the underlying structure of the inventory. An eigenvalue of 1 served as the criterion for retaining a factor, whereas a loading of .40 or greater was the standard for retaining a strategy within a particular factor. By means of the orthogonal varimax procedure, four factors emerged that we found to be theoretically and statistically justifiable.

Strategies within Factor 1 appear related to the active processing of the text (i.e., image, link to background knowledge, make predictions, use self-questioning, ignore noncritical words, and look for salient details). Factor 2 strategies apparently were those procedures students used to process the text at a more surface level (i.e., reread parts of the text, change reading rate, and use the context to determine meaning). Factor 3 seemed to have more to do with identifying the gist or main ideas contained in the passage (i.e., rehearse the main ideas, elaborate main ideas, reflect on the reading, and summarize mentally). What unified the strategies in Factor 4 was that they were virtually unused by students as they studied the passages at pretest (i.e., consult reference materials and look up vocabulary). These four factors explained 42% of the variance in reported strategy use at pretest. Important factors, such as prior knowledge and a willingness to engage in strategic processing, may also have contributed to students' reported strategy use.

On the inventory, students were directed to checkmark any strategy that they used to comprehend and remember the passage. When they finished, they placed an asterisk next to those strategies they considered most helpful. Students received 0 points for any unmarked strategy, 1 point for all checked strategies, and 2 points for those they asterisked (maximum score = 30). The Cronbach alpha for scores on the inventory was .54. As with the knowledge measure, we attribute this moderate reliability to low variability among participants at pretest. Also at pretest, a number of strategies were unused by the students, thereby constraining the range of scores.

Criterion Measure

As the external criterion on which we built our student profiles, we selected a domain-specific analogy task. This task consisted of eight classic or simple problems of the form A:B::C:D (e.g., Pavlov: classical conditioning::Skinner: *operant conditioning*). The content for these problems, as with other measures in our battery, was taken directly from material covered during the semester. We originally had conceived of the analogy task as one component of

the domain knowledge variable. However, subsequent review and analysis convinced us that this task would serve well as the outcome measure in this study for theoretical and statistical reasons.

From a theoretical standpoint, there is an ample literature indicating that successful performance on verbal analogy problems (Embretson, 1983; Goswami, 1992; Sternberg & Rifkin, 1979), including those specific to a domain of study (Alexander, Murphy, & Sperl, 1997; White & Alexander, 1986), require both conceptual knowledge and strategic capability. Thus, the use of an analogy task afforded us an outcome measure that encompassed two central dimensions within our profiles: domain knowledge and strategic processing. Further, on the basis of a confirmatory factor analysis by means of structural equation modeling (AMOS; Arbuckle, 1997), this analogy task was found to be a highly valid measure of domain-specific analogical reasoning (see Alexander, Murphy, & Kulikowich, 1998, for indices of fit).

The scoring of these items followed a categorization scheme that has been validated in prior studies by means of multidimensional scaling (Alexander, 1990) and item response theory (Alexander, Murphy, & Kulikowich, 1998). The interrater reliability for the scoring of the analogy task was .98. We, along with a doctoral student, determined this rate by independently coding the responses for 33 (10%) randomly selected participants. Because of the high level of agreement (Engelhard, 1996), the remaining items were scored by Murphy only. These response categories were hierarchically ordered on the basis of the knowledge and strategic capability each was assumed to convey (see Table 1). For example, Category 3 (Nondomain) errors covered student responses that fell outside of the target domain, educational psychology (e.g., Pavlov: classical conditioning::Skinner: *romantic conditioning*). Category 5 (Domain) errors, by comparison, demonstrated student's ability to produce a response from educational psychology (e.g., Pavlov: classical conditioning::Skinner: *reinforcement*).

Alexander, Murphy, and Kulikowich (1998) found not only that these response categories were hierarchical across respondents but also that students tended to display certain error patterns in their solutions. Because of the specificity at which student responses could be marked using this categorization scheme, we applied a partial-credit scoring model (Masters, 1982). That is, we awarded 1 point for a Nonresponse (Category 1) to 7 points for a Correct Response (Category 7). Thus, the highest possible score on the analogy task was 56.

Results and Discussion

Descriptive Data

Before submitting these data to cluster-analytic procedures, we first inspected the univariate distributions of all

Table 1
Response Categories Used in Scoring Domain-Specific Analogy Problems

Category	Description	Example
1	Nonresponse	Pavlov: classical conditioning::Skinner: ?
2	Repetition	Pavlov: classical conditioning::Skinner: <i>classical conditioning</i>
3	Nondomain Response	Pavlov: classical conditioning::Skinner: <i>romantic conditioning</i>
4	Structural Dependency	Pavlov: classical conditioning::Skinner: <i>neoclassical conditioning</i>
5	Domain Response	Pavlov: classical conditioning::Skinner: <i>reinforcement</i>
6	Target Variant	Pavlov: classical conditioning::Skinner: <i>operative conditioning</i>
7	Correct Response	Pavlov: classical conditioning::Skinner: <i>operant conditioning</i>

five variables, to ensure that they were approximately normal. The means and standard deviations for these variables at pretest and posttest are displayed in Table 2. Several patterns emerged from these data. First, students' scores on all variables increased significantly from pretest to posttest. Increases in knowledge, interest, strategy use, and recall might well be expected after a semester of formal instruction in educational psychology. However, there is no direct way to ascribe causality in this study. Still, in light of the 15-week delay between pretest and posttest, it is unlikely that these significant changes can be attributed simply to a test-retest effect (see Willson & Putnam, 1982, for an extended discussion of pretest sensitization effects).

Second, we found that the analogical reasoning variable displayed the greatest increase from pretest to posttest, $t(328) = 51.74, p < .0001$. Because measures of analogical reasoning are frequently used as indicators of intelligence or cognitive aptitude, due in part to strong test-retest reliabilities (e.g., Goswami, 1992; Klix, 1992; Sternberg, 1977), we posited that this significant change was attributable in some measure to the instruction students received. What was relevant about this instruction was that it focused not only on strategic processes (e.g., inferring, mapping) associated with analogical reasoning (e.g., Alexander, White, Haensly, & Crimmins-Jeanes, 1987; Sternberg, 1977) but also on domain concepts to which these processes were applied. Thus, by bringing their enhanced domain-specific knowledge and strategic processes to bear collectively on this task, students may have been able to register significant improvement from pretest to posttest. This trend in the data supported our choice of the analogy task as the external criterion for the cluster analysis.

Despite these recorded increases from pretest to posttest, students' scores on the cognitive and motivational measures, especially at pretest, were typically low to moderate. Only the interest ratings could be considered somewhat high at the outset of the course. The lowest relative scores occurred for the strategy use and recall measures. At pretest, students tended to rely on rereading and similar surface-level processing strategies (e.g., changing reading rate) to guide their studying. The result was an average of less than 4 idea units recalled out of a possible 85 or 93. We attributed this low score also to the task directive. Many recall tasks call on students to write down everything they can remember from a given passage. In this case, students were told to list the

"important" information from the passages. This may have limited the content students chose to record.

Not only did the average number of recalled idea units double at posttest but there were also quantitative and qualitative differences in students' strategy use. Specifically, the moderate rise in reported strategies was complemented by the fact that students marked more strategies that pertained to active processing or to identifying main ideas. In effect, students seemed to rely more on deeper processing strategies (e.g., summarizing) than surface-level ones (e.g., rereading) by the conclusion of the course. The preference for such strategies also seems in keeping with the nature of the recall task. Alexander et al. (in press) recently argued that such a quantitative-qualitative shift in learners' strategy use could be viewed as evidence of growth in strategic processing.

Cluster Analysis

As noted, we used cluster analysis to detect emergent profiles in the domain of educational psychology. Milligan and Cooper (1987) recommended a technique in which one variable, an external criterion, is used to validate the differences among profiles. We chose analogical reasoning as that external criterion for the reasons previously stated. With analogical reasoning as the external criterion and domain knowledge, interest, strategy use, and recall as the clustering measures, we examined students' pretest and posttest profiles. Specifically, we used Ward's minimum-variance hierarchical clustering technique (Everitt, 1993; Ward, 1963). In this analysis, differences in analogical reasoning ability served as an independent indicator of cluster-group distinctions. In effect, the relationship between cluster profiles and the external criterion variable functions as a measure of convergent validity. We viewed the pretest phase of this research as somewhat confirmatory, due to the availability of relevant cluster-analytic studies in the literature against which to compare our resulting profiles. However, our attempts to examine student profiles at posttest, and certainly our consideration of the role of formal instruction in contributing to such profiles, should be judged as rather exploratory.

Of the family of cluster-analytic techniques, we chose Ward's hierarchical clustering procedure because it is considered particularly effective in recovering the underlying

Table 2
Means and Standard Deviations of Scores on Measures at Pretest and Posttest

Variable	Pretest		Posttest		Max
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Analogical reasoning	19.40	7.87	44.23	4.99	56
Domain knowledge	12.66	2.26	17.18	2.43	24
Individual interest	126.68	20.15	145.95	18.39	182
Strategic use	7.80	3.71	9.07	3.80	30
Recall	3.88	2.56	8.58	3.64	85/93

Note. All differences in performance from pretest to posttest, as assessed by a *t* test for paired samples, were significant at $p < .001$, by the Bonferroni adjustment. Maximum (Max) for recall was 85 for knowledge passage and 93 for motivation passage.

structure of data (e.g., Atlas & Overall, 1994). Ward's hierarchical clustering procedure is an agglomerative method. That is to say, this procedure starts with every individual representing a unique cluster and then continues to examine distances between clusters in a stepwise fashion until all individuals (i.e., response vectors) have been grouped into one cluster. Distances are typically measures of dissimilarity between two objects. In this study, distance was a measure of dissimilarity between two individuals given their responses on the knowledge, interest, and recall measures.

Finally, we used dendrograms to aid our interpretation of the resulting clusters (Olson & Biolsi, 1991). In these dendrograms, the number of meaningful clusters is based on an identification of the largest gaps or distances between cluster groups. Two such gaps emerged in the dendrogram at pretest and three at posttest. These results suggest that three student profiles could statistically account for academic performance at the outset of the semester, whereas four distinct clusters were required to capture student performance at posttest. At pretest, the emerging clusters bore similarities to those we identified in the literature.

Descriptions of the Pretest Clusters

The means and standard deviations for the criterion and clustering variables for the three pretest clusters are shown in Table 3. To understand the differences among these clusters, we first conducted a multivariate analysis of variance (MANOVA). For this analysis, cluster group was the independent variable, and domain knowledge, interest, strategy use, and recall were the dependent variables. Results indicated a significant overall difference between clusters, $F(10, 642) = 54.71, p < .0001$. We followed this omnibus test with univariate analyses of variance (ANOVAs) for the four dependent variables and for the external criterion variable, analogical reasoning. The cluster groups were significantly different on all dependent measures, $F(2, 328) \geq 35.64, p < .00001, MSE \leq 333.10$. Performance on the analogical reasoning task also differed significantly for the clusters, $F(2, 328) = 4.44, p < .01, MSE = 60.65$.

Next, to obtain a better understanding of the significant univariate effects among the three cluster groups, we conducted a series of multiple comparisons by Fisher's

(1935) least significant difference (LSD) procedure. We chose Fisher's LSD procedure ($\alpha = .05$) because it exhibits higher statistical power when compared with other multiple comparison tests (e.g., Levin, Serlin, & Seaman, 1994). The results of these between-cluster comparisons are presented in Table 3 as subscript designations. We consider these differences in the general descriptions of the three clusters that follow. On the bases of their attributes, we refer to those three as the *Learning-Oriented*, *Strong-Knowledge*, and *Low-Profile* clusters.

Finally, we performed a discriminant function analysis, to validate the presumed multidimensional character of the clusters. This analysis would provide support for the notion that the emerging clusters arose from an interplay of cognitive and motivational variables (e.g., recall and interest) and could not be attributed to differences in a single construct (e.g., knowledge). For this discriminant function analysis, performance on the analogical reasoning, knowledge, interest, strategy use, and recall measures was used to predict group membership. Results demonstrated that overall group membership was accurately predicted for 84% of cases. The highest prediction accuracy occurred for the Strong-Knowledge group (88.2%), followed by the Low-Profile and the Learning-Oriented clusters (83% and 82.4%, respectively). As a last step in this analysis, we calculated Cohen's kappa, to determine the percentage of correct classification over and above what would be expected due to chance. We found a 70% improvement over chance. Collectively, these analyses provide strong support for the contention that the three clusters that emerged at pretest were not only statistically unique but also multidimensional in nature.

Learning-Oriented cluster. What seemed to distinguish the Learning-Oriented cluster ($n = 108$) was that students displayed an initially moderate level of domain knowledge with a high level of domain-specific interest. They appeared to engage in the most effortful processing of the knowledge and motivation passages also and, subsequently, remembered the most from those passages relative to the other clusters. In addition, they recorded one of the highest means on the analogical reasoning task at pretest. Thus, it would appear that those in this particular cluster were quite interested in learning the content of educational psychology

Table 3
Means and Standard Deviations of Scores on Measures
for the Three Student Clusters at Pretest

Variable	Cluster						Max
	Learning Oriented		Strong Knowledge		Low Profile		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Analogical reasoning	20.64 _a	7.99	20.53 _a	8.21	18.03 _b	7.45	56
Domain knowledge	12.82 _b	1.90	14.84 _a	1.78	11.58 _c	1.94	24
Individual interest	135.85 _a	20.66	111.92 _c	10.97	126.76 _b	19.11	182
Strategic use	10.42 _a	3.99	7.49 _b	2.88	6.09 _c	2.64	30
Recall	5.40 _n	3.13	4.21 _b	1.80	2.67 _c	1.64	85/93

Note. Learning Oriented $n = 108$; Strong Knowledge $n = 68$; Low Profile $n = 153$. Maximum (Max) for recall was 85 for knowledge passage and 93 for motivation passage. Means with different subscripts differ significantly at $\alpha = .05$ by the Fisher least significant difference test.

and were willing to exert the strategic effort required for such learning. It is for this reason that we refer to them as *learning oriented*; that is, reflecting a learning or mastery goal orientation (e.g., Dweck & Leggett, 1988; Meece & Holt, 1993). Overall, this cluster is closest to the high-knowledge-high-interest group, which led performance in the Alexander et al. (1995) experiments, or to the highly successful principle-based reasoners in the Renkl (1997) research.

Strong-Knowledge cluster. By comparison, students in the Strong-Knowledge group ($n = 68$) apparently began the course with more relevant knowledge than those in other clusters. These students tended also to rely on that existing knowledge to guide their performance. These students' interest in educational psychology, by comparison, was the lowest of the three clusters, and their strategic efforts, as well as their passage recalls, were rather moderate. Comparable profiles have been reported by others (e.g., de Jong & Ferguson-Hessler, 1986; Recker & Pirolli, 1995; Renkl 1997). As was true in these prior studies, students fitting this profile were still able to perform successfully on the outcome measures at the beginning of the term. It would seem, therefore, that a relevant body of subject-matter knowledge could potentially compensate for limited interest and moderate effort after meaning, at least in the initial stages of domain learning (Alexander, 1997).

Low-Profile cluster. The largest cluster at pretest, the Low-Profile group ($n = 153$), reported only a moderate interest in the domain of educational psychology and were seemingly the least informed in this domain. Further, these students used the least number of strategies and remembered the least of the three clusters. Likewise, they were significantly below the other two clusters on the criterion task. These relatively low scores in analogical reasoning seem understandable in the light of students' performance on the associated knowledge and strategy tasks. This Low-Profile group is analogous to the low-profile cluster in the Alexander et al. (1995) study and to the passive and superficial processors described by Recker and Pirolli (1995). It is understandable that such a large portion of students would fall into this Low-Profile group before formal instruction in educational psychology or in strategic processing.

Descriptions of the Posttest Clusters

After approximately 15 weeks of explicit instruction in the domain, along with the discussion and practice of various learning and studying strategies, our participants fell into four distinct clusters. As we did at pretest, we first examined the differences among these groups by means of MANOVA, with cluster group as the independent variable and the knowledge, interest, strategy use, and recall measures as dependent variables. The outcome indicated an overall difference among the clusters, $F(12, 962) = 46.93$, $p < .0001$. The univariate ANOVAs we conducted showed that the four clusters varied significantly with regard to their domain knowledge, interest, strategy use, and passage recall, $F(3, 328) \geq 23.41$, $p < .00001$, $MSE \leq 179.40$. The cluster groups' performance on the domain-specific analogy problems, the external criterion variable, also differed significantly, $F(3, 328) = 4.86$, $p < .003$, $MSE = 24.03$.

Again, we used Fisher's LSD procedure to explore the between-cluster differences on these five variables. For this analysis, we used Hayter's adaptation (Hayter, 1986) of the LSD procedure because there were four levels of the independent variable, cluster group. In such cases, the Fisher-Hayter method provides more control for familywise Type I error rates (Seaman, Levin, & Serlin, 1991). The outcomes of these multiple comparisons are represented in Table 4 as subscripts. We used these data also to generate descriptions of the four resulting profiles, which we labeled the *Learning-Oriented*, *Strong-Knowledge*, *Effortful-Processor*, and *Nonstrategic-Reader* clusters. The first two labels were retained to convey profiles similar to those at pretest.

Finally, we performed discriminant function analysis, to validate the multidimensional character of the four clusters, with analogical reasoning, knowledge, interest, strategy use, and recall as the predictor variables. Overall, we found that cluster membership was accurately predicted for 84% of the participants. As at pretest, prediction accuracy was the best for the Strong-Knowledge group (89.7%). This was followed by predictions for the Nonstrategic-Reader cluster (87.5%), the Effortful-Processor cluster (79.8%), and the Learning-Oriented group (79.8%). When we calculated Cohen's kappa, we ascertained that this particular configura-

Table 4
Means and Standard Deviations of Scores on Measures for the Four Student Clusters at Posttest

Variable	Cluster								Max
	Learning Oriented		Strong Knowledge		Effortful Processor		Nonstrategic Reader		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Analogical reasoning	45.49	4.76	45.46 _a	4.75	43.55 _b	4.79	43.28 _b	5.12	56
Domain knowledge	18.17	2.28	18.82 _a	2.46	15.96 _c	1.94	16.65 _b	2.23	24
Individual interest	156.52	12.36	113.38 _c	17.72	145.67 _b	14.66	148.43 _b	11.61	182
Strategic use	9.78	3.30	7.77 _c	3.13	12.66 _a	3.61	6.67 _d	2.22	30
Recall	11.79	2.99	8.54 _b	3.87	8.36 _b	2.48	6.22 _c	2.66	85/93

Note. Learning Oriented $n = 94$; Strong Knowledge $n = 39$; Effortful Processors $n = 76$; Nonstrategic Readers $n = 120$. Maximum (Max) for recall was 85 for knowledge passage and 93 for motivation passage. Means with different subscripts differ significantly at $\alpha = .05$ by the Fisher least significant difference test.

tion of cluster membership represented a 75% improvement over chance.

Learning-Oriented cluster. Once again, students in the Learning-Oriented cluster ($n = 94$) performed well on all the cognitive and motivational tasks. It was only on the strategy use measure that the learning-oriented students did not record one of the highest cluster means. Note that the domain knowledge of those in this cluster was virtually equivalent to that of students in the Strong-Knowledge group at posttest. Thus, it would seem that the goal of learning was achieved for those in this cluster. Even at the conclusion of the semester, students in the Learning-Oriented group retained their high interest in the domain, were quite effortful in their studying, and were effective in their recall of important passage information. They used fewer strategies than did students in the Effortful-Processor group, yet they documented significantly more strategies than the other two clusters nonetheless. They also tended to rely heavily on deep-processing over surface-level strategies in their studying. Finally, the number of students within this cluster was rather consistent from pretest to posttest.

Strong-Knowledge cluster. The profile for students in the Strong-Knowledge group was also relatively stable from pretest to posttest. However, there was a noticeable decrease in the number of students associated with this particular cluster from the outset to the conclusion of the term (68 vs. 39, respectively). As before, their low interest in the domain and their minimum strategic output during studying characterized these students. Specifically for this group, interest was low at pretest and remained stable throughout the course. Nonetheless, their foundation of knowledge allowed them to perform well on the passage-recall task, perhaps by reducing the number of comprehension problems they encountered during processing. Moreover, the Strong-Knowledge students performed as well as those in the Learning-Oriented group on the knowledge measure and on the analogy task. Several researchers have suggested that knowledge and individual interest are significantly correlated, in that those with the greatest knowledge are likewise personally invested in the domain (Alexander, 1997; Renninger, 1992). For most students in this study, that association between knowledge and interest was supported. However, this was not the case for this particular group of students, who apparently did pretty well in a domain they cared relatively little about. Rather, the initial interest level for this group remained relatively unchanged.

Effortful-Processor cluster. What is quite interesting is what became of students who initially formed the Low-Profile cluster at pretest. Unlike the clusterings that emerged at pretest, there was no profile group that was consistently low on almost all cognitive and motivational dimensions. Rather, it appears that those 153 students in the Low-Profile group, along with others from the remaining clusters, re-formed into two contrasting profiles. First, there were those students ($n = 76$) who seemingly struggled very hard to learn the content, as indicated by their strategy use, and who even maintained a moderate level of interest in the domain. Yet, when it came to building a base of domain knowledge, these students were among the least successful

learners in the class. Thus, we refer to these students as *effortful processors*. The high level of strategic effort might help to explain how this group, with the lowest average on the domain knowledge measure, was still able to perform moderately well on both the recall and analogy tasks.

Perhaps these effortful processors learned to compensate for their limited domain knowledge by working more effortfully and more strategically. It is also possible that this group of students is simply less skilled at taking declarative knowledge measures, such as the domain knowledge task in this study. The profile for this group of students deviates from those in the literature. As evidenced from the labels attached to unsuccessful learners in prior studies (e.g., *passive* or *superficial*), low performance is attributed either to disinterest or to an unwillingness to engage. This does not appear to be the case for this segment of our student population. Instead, there was a noticeable disparity between their level of effort and knowledge gains.

Nonstrategic-Reader cluster. The final cluster to emerge at posttest is what we call the *Nonstrategic-Reader* profile ($n = 120$). As this name implies, this cluster distinguished itself by exerting the least amount of strategic effort during studying, as documented by their reported strategy use. Understandably, it was also this group who, on average, remembered the least from the knowledge and motivation passages that they read. Still, this low strategy and recall performance was balanced by moderate outcomes on the domain knowledge and analogical reasoning tasks. Additionally, those in the Nonstrategic-Reader cluster claimed to be at least moderately interested in the domain of educational psychology.

This contrasting profile leads us to question whether these students were somehow limited in their ability to learn from exposition. On the other hand, are those in this particular cluster the superficial learners that Renkl (1997) described in his research? That is, it could be that these students' reported interest in the domain did not translate into a willingness to study and recall domain-related information presented within demanding exposition. The fact that this cluster contained the largest number of students at the conclusion of the semester definitely gives us pause. This profile also has implications for students' academic development, considering that learning still depends heavily on the individuals' ability to acquire information from written text (Alexander & Knight, 1993; Alexander & Kulikowich, 1991). In a related cluster-analytic study, Lorch, Lorch, and Klusewitz (1993) raised similar concerns about the number of undergraduates who seemingly lacked the strategic ability required to comprehend demanding domain exposition.

Student Changes From Pretest to Posttest

We were interested also in the shifts that occurred in individual participants who constituted the various clusters at pretest and posttest. For instance, what happened to those students who began the semester with a high level of interest, a moderate level of knowledge, and a willingness and ability to study and learn from expository text? Did those students who constituted the Learning-Oriented clus-

ter at pretest maintain their commitment to learning, and were they able to translate a desire to learn into an ability to learn? If this were the case, then we would expect to see a large portion of those students in the pretest Learning-Oriented cluster as members of the Learning-Oriented cluster at posttest. Also, we were interested in those students who fell into the Low-Profile cluster at pretest. Because no clusters were similarly low on all cognitive and motivational dimensions at posttest, it is evident that change did occur during the course of the semester. But what form did that change take?

To address these questions, we tracked the movement of all 329 students from pretest to posttest in terms of their cluster membership. The results of this tracking are displayed in Figure 1. As this discussion unfolds, it is important to remember that this analysis should be viewed as exploratory. We also ran chi-square tests, to ascertain whether these movements were beyond the chance level, which they were, $\chi^2(3, 329) \geq 15.65, p \leq .013$. What we determined was that almost 40% of the students characterized as Learning Oriented at pretest were Learning Oriented at posttest. Approximately one quarter of Learning-Oriented students at pretest, however, found their way into either the Effortful-Processor cluster (26%) or the Nonstrategic-Reader group (26%). This suggests that a substantial portion of those who began with laudable goals again demonstrated a disposition to work hard at learning. Yet they were unable to convert this effort into a high level of subject-matter knowledge. On the other hand, an equivalent number of those students with high interest and initial effortfulness found their way into the cluster distinguished by their inability or unwillingness to learn from demanding text.

By and large, the participants who constituted the Low-Profile cluster at pretest tended to find their way into the Nonstrategic-Reader group (40%), followed by the Learning-Oriented (25%) and the Effortful Processor (24%) clusters. Consequently, it appears that all these low-low students in the Low-Profile cluster garnered something from the semester of instruction. The largest percentage acquired a base of educational psychology knowledge and an increased interest in the domain but not the ability or willingness to gain information from demanding text (Nonstrategic Reader). Others from this cluster ended the semester with relatively high interest in the domain that was reflected also in their studying behavior (Effortful Processor). Still, their ability to acquire a base of domain knowledge comparable to their peers was limited. One quarter of those with a consistently low profile at the outset of the term turned in a consistently high performance at its conclusion (Learning Oriented), which is encouraging indeed.

Finally, the students we described as Strong Knowledge at pretest were, in many ways, the least stable cluster members. Only 18% of those high-knowledge students retained that label at posttest. The greatest percentage of this group (46%) was ranked among those who had substantial knowledge and some interest but not the ability or the desire to learn from text (Nonstrategic Reader). Thus, it would seem that beginning an introductory course in a domain with relatively more

knowledge than one's peers is not a clear path to high knowledge, personal investment, or strategic capability.

Conclusion and Implications

Three questions, dealing with the nature, consistency, and malleability of academic profiles, guided this investigation. The principal limitation in addressing these questions rests in our inability to make any definitive statements about the effects of the course instruction on the clusters we characterized or the shifts in profiles that we documented. It is logical to assume that this shared experience in the form of explicit instruction in domain-related concepts and general strategies would have some impact on students' knowledge, interest, and strategic processing. This seems particularly true because our measures were specific to the domain of educational psychology. However, the absence of controls constrains our ability to ascribe causality. Thus, our results must be judged as exploratory and require later confirmation or refutation. Likewise, we recognize that the validity of cluster analysis relies on the quality of the measures used to build the profiles, as well as the identification of a psychometrically sound criterion against which these profiles can be evaluated (Milligan & Cooper, 1987). Although we have labored to produce sound indicators of domain knowledge, interest, and strategic processing, alternative measures or the choice of other constructs may have resulted in varied groupings. Despite these potential shortcomings, we believe that this study contributes to the literature on learning and instruction in several important ways. Particularly noteworthy is a glimpse into the character of intraindividual differences (Ackerman, in press).

First, as it pertains to the nature of academic profiles, this study amplified that success in instructional settings entails the orchestration of cognitive and noncognitive factors. Knowledge, interest, or strategic ability alone did not ensure optimal academic performance. The integration of all three is what marked maximal performance. Although an adequate base of domain knowledge, of all the variables that we investigated, did contribute to higher performance, its power weakened over time without a concomitant interest in the domain and without the capability to learn from formidable exposition.

Second, we were struck by the relative consistency in the profiles we uncovered with those described in the literature. The fact that the participants and the domain in the present study varied markedly from those researched by others makes such consistency even more meaningful. Thus, this consistency adds greater credence to the patterns observed in this study. What those patterns reveal is that there generally appears to be two paths to academic success, at least in the initial stages of formal learning. The first, less superlative road is one in which learners, who have somehow acquired a base of relevant knowledge, rely extensively on that knowledge to propel their performance. The second path, and one that results in even higher levels of overall achievement, combines knowledge with a personal investment in the learning process, as evidenced by strong interest in the content and strategic effort. Initially, lower levels of knowl-

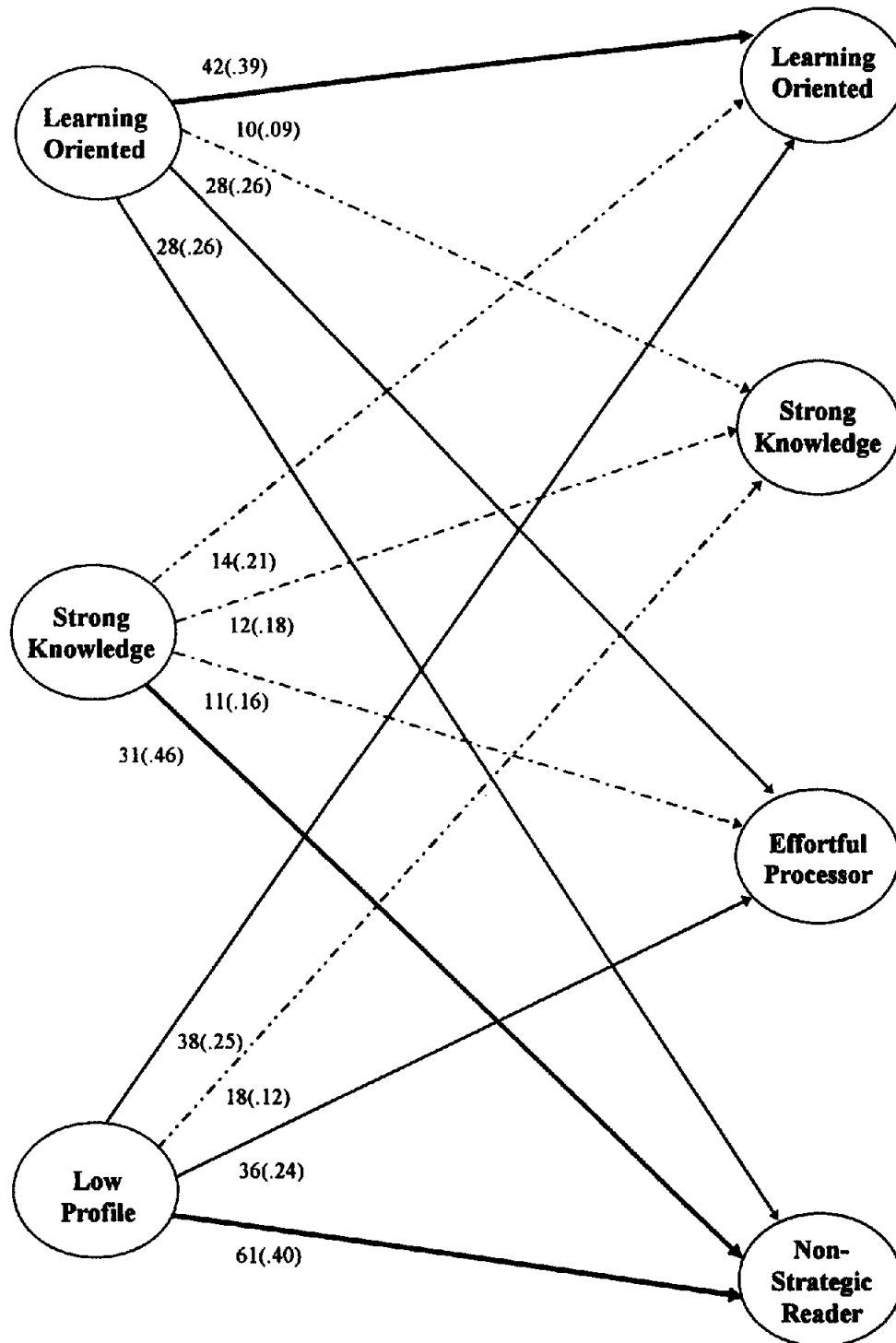


Figure 1. Representation of changes in cluster membership from pretest to posttest, with number (and relative proportion) of students moving from a specific pretest cluster to a particular posttest cluster.

edge, interest, and strategic ability distinguish the less successful learner.

Perhaps the most significant contribution of this research was our opportunity to explore the fluidity or malleability of

these learner profiles as formal instruction in the target domain unfolded. We were in the unique position to examine the changes in learner attributes after 15 weeks of relevant instruction and to ponder the circumstances that might

stimulate or stifle students' academic development. We had hypothesized that a semester of instruction in the concepts of educational psychology and in general strategies would influence positively the knowledge, interest, and strategic processes that students initially displayed. Moreover, we anticipated that the diversities that the students revealed would be magnified by this shared educational experience. This relates to the statistical notion of a fan-spread effect (Bryk & Weisberg, 1977).

Pretest to posttest comparisons supported these hypotheses. By the conclusion of the term, the undergraduates (a) demonstrated significant growth in domain knowledge, (b) were more personally interested in that domain, (c) engaged in deeper processing of text, and (d) recalled more of what they read. Yet these mean differences were not equally distributed among all members of this classroom community. Rather, noticeable changes seemed tied to the students' initial strengths and weaknesses, as mirrored in their academic profiles. Stanovich (1986) would explain this in terms of the "Matthew effect," in which relevant learning experiences exacerbate differences between the academic haves and have-nots. Thus, the three profiles that emerged at pretest gave way to four groupings, with students migrating to a certain extent to these new academic configurations.

Still, the "high road" to academic success remained one in which knowledge, interest, and strategic ability were high. It is encouraging that many students entered the classroom with just such a learning profile. Others who started out in the Strong-Knowledge or Low-Profile clusters found their way into this high-knowledge-high-interest-high-strategic-processing cluster. Despite this encouraging trend, two less positive groupings emerged by the end of the semester. There were those students whose high interest and strategic efforts did not convert into comparable domain knowledge or analogical reasoning performance. Perhaps with additional instructional support, a number of these students may have realized the academic success of the learning-oriented students.

There was also a large number of students in this introductory course who fit the profile of a nonstrategic reader. Whether this tendency is attributable to an inability to read strategically or to a reluctance to exert the necessary cognitive or metacognitive energy, the fact remains that many students did not perform well when presented with lengthy and challenging text (Lorch et al., 1993). When we factor in the amount of academic content students confront in written form during their educational career (Alexander & Kulikowich, 1991), the magnitude of this problem is unveiled. Clearly the assistance these nonstrategic readers received during this one semester was inadequate to overcome habits or deficiencies that possibly were years in the making. On the basis of this pattern, educators may want to reconsider the attention that they pay to text-processing strategies during the early years of students' academic development. At the very least, those students who are willing to exert the effort to learn from text may be better equipped to succeed at that task.

For us, the Strong-Knowledge profile was the most perplexing group of students. This particular cluster of

students was able to perform adequately on the cognitive tasks presented them, but they conveyed a passivity or lack of personal investment in the learning process. In effect, their minds were in their learning, but not their spirits. Perhaps it is unrealistic to assume that all students will become interested in the content they are learning. After all, these students are acquiring concepts central to the domain, even in spite of their relative disinterest or lack of strategic investment. However, this contingent of students defies the general perception in the literature and trend in the data that knowledge and interest should be significantly and positively correlated. Alexander (1997) would argue that these students' subsequent academic development in the domain of educational psychology would be thwarted without an ensuing rise in individual interest and strategic investment. Yet this contention remains to be verified for this particular group of students.

One of our goals in this investigation was to consider the instructional implications of these profiles. From a practical standpoint, this investigation allows us to look within a large community of learners and observe how individuals cluster together on the basis of their domain knowledge, individual interest, and strategic processing: As Dewey phrased it.

The teacher's problem is thus twofold. On the one side, he needs . . . to be a student of individual traits and habits; on the other side, he needs to be a student of the conditions that modify for better or worse the directions in which individual powers habitually express themselves. (Dewey, 1910/1993, p. 46)

Of course, without further investigation, it remains unclear whether knowledge of students' academic profiles can assist educators in orchestrating learning environments that are more apt to address such patterns of individual differences. For instance, should students with little knowledge of and little interest in the domain be presented with opportunities distinct or varied from the opportunities afforded those with higher knowledge, interest, and a commitment to gain competence in that domain? Given that the intraindividual profile manifested by learning-oriented students translated into optimal performance on our various tasks, we might also ask what conditions spark such an orientation in students? Is it possible to create learning environments that convert more passive or reluctant learners into those who are concerned with mastering a particular domain? What must be done to compose environments that maintain students' initial levels of interest and commitment (Mitchell, 1993)? Because this investigation must be viewed as exploratory with regard to the transformations in students' profiles, we can only pose such critical questions for future study. Still, in our judgment, such empirical questions are indeed worthy of exploration.

References

- Ackerman, P. L. (in press). Traits and knowledge as determinants of learning and individual differences: Putting it all together. In P. L. Ackerman, P. C. Kyllonen, & R. D. Roberts (Eds.), *The future of learning and individual-differences research: Pro-*

- cesses, traits, and content. Washington, DC: American Psychological Association.
- Ackerman, P. L., Kanfer, R., & Goff, M. (1995). Cognitive and noncognitive determinants and consequences of complex skill acquisition. *Journal of Experimental Psychology: Applied*, 1, 270-304.
- Ackerman, P. L., Kyllonen, P. C., & Roberts, R. D. (in press). *The future of learning and individual-differences research: Processes, traits, and content*. Washington, DC: American Psychological Association.
- Alexander, P. A. (1990, April). *Categorizing learner responses on domain-specific analogy tests: A case for error analysis*. Paper presented at the annual meeting of the American Educational Research Association, Boston.
- Alexander, P. A. (1992). Domain knowledge: Evolving themes and emerging concerns. *Educational Psychologist*, 27, 33-51.
- Alexander, P. A. (1997). Mapping the multidimensional nature of domain learning: The interplay of cognitive, motivational, and strategic forces. In M. L. Maehr & P. R. Pintrich (Eds.), *Advances in motivation and achievement* (Vol. 10, pp. 213-250). Greenwich, CT: JAI Press.
- Alexander, P. A., Graham, S., & Harris, K. (in press). A perspective on strategy research: Progress and prospects. *Educational Psychology Review*.
- Alexander, P. A., Jetton, T. L., & Kulikowich, J. M. (1995). Interrelationship of knowledge, interest, and recall: Assessing a model of domain learning. *Journal of Educational Psychology*, 87, 559-575.
- Alexander, P. A., & Knight, S. L. (1993). Dimensions of the interplay between learning and teaching. *Educational Forum*, 57, 232-245.
- Alexander, P. A., & Kulikowich, J. M. (1991). Domain-specific and strategic knowledge as predictors of expository text comprehension. *Journal of Reading Behavior*, 23, 165-190.
- Alexander, P. A., Kulikowich, J. M., & Schulze, S. K. (1994). How subject-matter knowledge affects recall and interest on the comprehension of scientific exposition. *American Educational Research Journal*, 31, 313-337.
- Alexander, P. A., & Murphy, P. K. (1998). The research base for APA's learner-centered principles. In N. Lambert & B. L. McCombs (Eds.), *Issues in school reform: A sampler of psychological perspectives on learner-centered schools* (pp. 25-60). Washington, DC: American Psychological Association.
- Alexander, P. A., Murphy, P. K., & Kulikowich, J. M. (1998). What responses to domain-specific analogy problems reveal about emerging competence: A new perspective on an old acquaintance. *Journal of Educational Psychology*, 90, 397-406.
- Alexander, P. A., Murphy, P. K., & Sperl, C. T. (1997, March). *Measuring emerging competence through analogies: A new perspective on an old acquaintance*. Paper presented at the annual meeting of the American Educational Research Association, Chicago.
- Alexander, P. A., Murphy, P. K., & Woods, B. S. (1996). Of squalls and fathoms: Navigating the seas of educational innovation. *Educational Researcher*, 25, 31-36, 39.
- Alexander, P. A., Murphy, P. K., Woods, B. S., Duhon, K. E., & Parker, D. (1997). College instruction and concomitant changes in students knowledge, interest, and strategy use: A study of domain learning. *Contemporary Educational Psychology*, 22, 125-146.
- Alexander, P. A., Schallert, D. L., & Hare, V. C. (1991). Coming to terms: How researchers in learning and literacy talk about knowledge. *Review of Educational Research*, 61, 315-343.
- Alexander, P. A., White, C. S., Haensly, P. A., & Crimmins-Jeanes, M. (1987). Training in analogical reasoning. *American Educational Research Journal*, 24, 387-404.
- American Psychological Association, Presidential Task Force on Psychology in Education. (1993). *Learner-centered psychological principles: Guidelines for school redesign and reform*. Washington, DC: Author.
- Arbuckle, J. L. (1997). *Amos users' guide version 3.6*. Chicago: SmallWaters.
- Atlas, R. S., & Overall, J. E. (1994). Comparative evaluation of two superior stopping rules for hierarchical cluster analysis. *Psychometrika*, 59, 581-591.
- Bryk, A. S., & Weisberg, H. I. (1977). Use of the nonequivalent control group design when subjects are growing. *Psychological Bulletin*, 85, 950-962.
- Chi, M. T. H., Bassok, M., Lewis, M. W., Reimann, P., & Glaser, R. (1989). Self-explanations: How students study and use examples in learning to solve problems. *Cognitive Science*, 18, 145-182.
- de Jong, T., & Ferguson-Hessler, M. G. M. (1986). Cognitive structure of good and poor novice problem solvers in physics. *Journal of Educational Psychology*, 78, 279-288.
- de Jong, T., & Ferguson-Hessler, M. G. M. (1991). Knowledge of problem situations in physics: A comparison of good and poor novice problem solvers. *Learning and Instruction*, 1, 289-302.
- Dewey, J. (1993). *How we think*. Buffalo, NY: Prometheus Books. Original work published 1910.
- Dweck, C. S., & Leggett, E. L. (1988). A social-cognitive approach to motivation and personality. *Psychological Review*, 95, 256-273.
- Embretson, S. E. (1983). A general latent trait model for response processes. *Psychometrika*, 49, 175-186.
- Engelhard, G., Jr. (1996). Evaluating rater accuracy in performance assessments. *Journal of Educational Measurement*, 33, 56-70.
- Everitt, B. S. (1993). *Cluster analysis* (3rd ed.). New York: Halsted Press.
- Fisher, R. A. (1935). *The design of experiments*. New York: Hafner.
- Gelman, R., & Greeno, J. G. (1989). On the nature of competence: Principles for understanding in a domain. In L. B. Resnick (Ed.), *Knowing, learning, and instruction: Essays in honor of Robert Glaser* (pp. 125-186). Hillsdale, NJ: Erlbaum.
- Goetz, E. T., Alexander, P. A., & Ash, M. (1992). *Educational psychology: A classroom perspective*. Columbus, OH: Charles E. Merrill.
- Goswami, U. (1992). *Analogical reasoning in children*. Hove, England: Erlbaum.
- Guthrie, J. T., & Wigfield, A. (Eds.). (1997). *Reading engagement: Motivating readers through integrated instruction*. Newark, DE: International Reading Association.
- Hayter, A. J. (1986). The maximum familywise error rate of Fisher's Least Significant Difference Test. *Journal of the American Statistical Association*, 81, 1000-1004.
- Hidi, S. (1990). Interest and its contribution as a mental resource for learning. *Review of Educational Research*, 60, 549-571.
- Holyoak, K. L. (1991). Symbolic connectionism: Toward third-generation theories of expertise. In K. A. Ericsson & J. Smith (Eds.), *Toward a general theory of expertise: Prospects and limits* (pp. 301-335). New York: Cambridge University Press.
- Klix, F. (1992). Higher order learning mechanisms in knowledge domain. *Zeitschrift für Psychologie*, 200, 91-103.
- Knight, S. L., & Alexander, P. A. (1994, July). *Investigating bilingual readers strategic processing of English and Spanish text*. Paper presented at the 23rd International Congress of Applied Psychology, Madrid, Spain.
- Levin, J. R., Serlin, R. C., & Seaman, M. A. (1994). A controlled, powerful multiple-comparison strategy for several situations. *Psychological Bulletin*, 115, 153-159.

- Lorch, R. F., Lorch, E. P., & Klusewitz, M. A. (1993). College students' conditional knowledge about reading. *Journal of Educational Psychology*, 85, 239-252.
- Masters, G. M. (1982). A Rasch model for partial credit scoring. *Psychometrika*, 47, 149-174.
- Meece, J. L., & Holt, K. (1993). A pattern analysis of students achievement goals. *Journal of Educational Psychology*, 85, 582-590.
- Milligan, G. W., & Cooper, M. C. (1987). Methodology review: Clustering methods. *Applied Psychological Measurement*, 11, 329-354.
- Mitchell, M. (1993). Situational interest: Its multifaceted structure in the secondary school mathematics classroom. *Journal of Educational Psychology*, 85, 424-436.
- Olson, J. R., & Biolsi, K. J. (1991). Techniques for representing expert knowledge. In K. A. Ericsson & J. Smith (Eds.), *Toward a general theory of expertise: Prospects and limits* (pp. 240-285). Cambridge, England: Cambridge University Press.
- Pichert, J., & Anderson, R. C. (1977). Taking different perspectives on a story. *Journal of Educational Psychology*, 69, 309-315.
- Pirolli, P., & Recker, M. (1994). Learning strategies and transfer in the domain of programming. *Cognition and Instruction*, 12, 235-275.
- Pressley, M., Borkowski, J., & Schneider, W. (1989). Good information processing: What is it and what education can do to promote it. *International Journal of Educational Research*, 13, 857-867.
- Pressley, M., Wood, E., Woloshyn, V. E., Martin, V., King, A., & Menke, D. (1992). Encouraging mindful use of prior knowledge: Attempting to construct explanatory answers facilitates learning. *Educational Psychologist*, 27, 91-109.
- Recker, M. M., & Pirolli, P. (1995). Modeling individual differences in students' learning strategies. *Journal of the Learning Sciences*, 4, 1-38.
- Renkl, A. (1997). Learning from worked-out examples: A study on individual differences. *Cognitive Science*, 21, 1-29.
- Renninger, K. A. (1992). Individual interest and development: Implications for theory and practice. In K. A. Renninger, S. Hidi, & A. Krapp, (Eds.), *The role of interest in learning and development* (pp. 361-395). Hillsdale, NJ: Erlbaum.
- Schiefele, U. (1991). Interest, learning, and motivation. *Educational Psychologist*, 26, 229-323.
- Seaman, M. A., Levin, J. R., & Serlin, R. C. (1991). New developments in pairwise multiple comparisons: Some powerful practicable procedures. *Psychological Bulletin*, 110, 577-586.
- Spiro, R. J., Coulson, R. L., Feltovich, P. J., & Anderson, D. K. (1994). Cognitive flexibility theory: Advance knowledge acquisition in ill-structured domains. In R. B. Ruddell, M. R. Ruddell, & H. Singer (Eds.), *Theoretical models and processes of reading* (pp. 602-615). Newark, DE: International Reading Association.
- Spiro, R. J., & Jehng, J. C. (1990). Cognitive flexibility and hypertext: Theory and technology for the nonlinear and multidimensional traversal of complex subject matter. In D. Nix & R. J. Spiro (Eds.), *Cognition, education, and multimedia* (pp. 163-205). Hillsdale, NJ: Erlbaum.
- Stanovich, K. E. (1986). Matthew effects in reading: Some consequences of individual differences in the acquisition of literacy. *Reading Research Quarterly*, 21, 360-407.
- Sternberg, R. J. (1977). *Intelligence, information processing, and analogical reasoning: The componential analysis of human abilities*. Hillsdale, NJ: Erlbaum.
- Sternberg, R. J., & Rifkin, B. (1979). The development of analogical reasoning processes. *Journal of Experimental Child Psychology*, 27, 195-232.
- Ward, J. H. (1963). Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58, 236-244.
- Weinstein, C. E., & Mayer, R. E. (1986). The teaching of learning strategies. In M. C. Wittrock (Ed.), *Handbook of research on teaching* (3rd ed., pp. 315-327). New York: Macmillan.
- White, C. S., & Alexander, P. A. (1986). Effects of training on four-year-olds' ability to solve geometric analogy problems. *Cognition and Instruction*, 3, 261-268.
- Willson, V. L., & Putnam, R. R. (1982). A meta-analysis of pretest sensitization effects in experimental design. *American Educational Research Journal*, 19, 245-258.

Received March 3, 1997

Revision received January 10, 1998

Accepted January 12, 1998 ■